

PHYS-467 Assignment 2

Pietro Giugiaro

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1 Question 3

The posterior is defined as

$$P(s|Y, \rho, p_E) = \frac{1}{Z} P(s) P(Y|s, \rho, p_E)$$

where the prior $P(s)$ and the likelihood $P(Y|s, \rho, p_E)$ are respectively:

$$P(s) = \prod_i \frac{1}{2} [\delta(s_i = 1) + \delta(s_i = -1)] = \frac{1}{2^N} \prod_i \mathbf{1}(s_i = \pm 1)$$

$$\begin{aligned} P(Y|s, \rho, p_E) &= \prod_{i < j} P(Y_{ij}|s, \rho, p_E) = \\ &= \prod_{i < j} [p_E(1 - \rho) \mathbf{1}(Y_{ij} = s_i s_j) + p_E \rho \mathbf{1}(Y_{ij} = -s_i s_j) + (1 - p_E) \mathbf{1}(Y_{ij} = 0)] = \\ &= [p_E(1 - \rho)]^{\sum_{i < j} \mathbf{1}(Y_{ij} = s_i s_j)} (p_E \rho)^{\sum_{i < j} \mathbf{1}(Y_{ij} = -s_i s_j)} (1 - p_E)^{\sum_{i < j} \mathbf{1}(Y_{ij} = 0)} \end{aligned}$$

Now it is possible to compute the marginal of the likelihood by integrating over p_E the joint probability $P(Y, p_E|s, \rho) = P(Y|s, \rho, p_E) P(p_E)$:

$$P(Y|s, \rho) = \int_0^1 P(Y|s, \rho, p_E) P(p_E) dp_E$$

Computing the integral one obtains

$$\begin{aligned} P(Y|s, \rho) &= \int_0^1 P(Y|s, \rho, p_E) P(p_E) dp_E = \\ &= \int_0^1 p_E [p_E(1 - \rho)]^{\sum_{i < j} \mathbf{1}(Y_{ij} = s_i s_j)} (p_E \rho)^{\sum_{i < j} \mathbf{1}(Y_{ij} = -s_i s_j)} (1 - p_E)^{\sum_{i < j} \mathbf{1}(Y_{ij} = 0)} dp_E = \\ &= (1 - \rho)^{\sum_{i < j} \mathbf{1}(Y_{ij} = s_i s_j)} \rho^{\sum_{i < j} \mathbf{1}(Y_{ij} = -s_i s_j)} \int_0^1 f(p_E) dp_E = \\ &\propto (1 - \rho)^{\sum_{i < j} \mathbf{1}(Y_{ij} = s_i s_j)} \rho^{\sum_{i < j} \mathbf{1}(Y_{ij} = -s_i s_j)} \end{aligned}$$

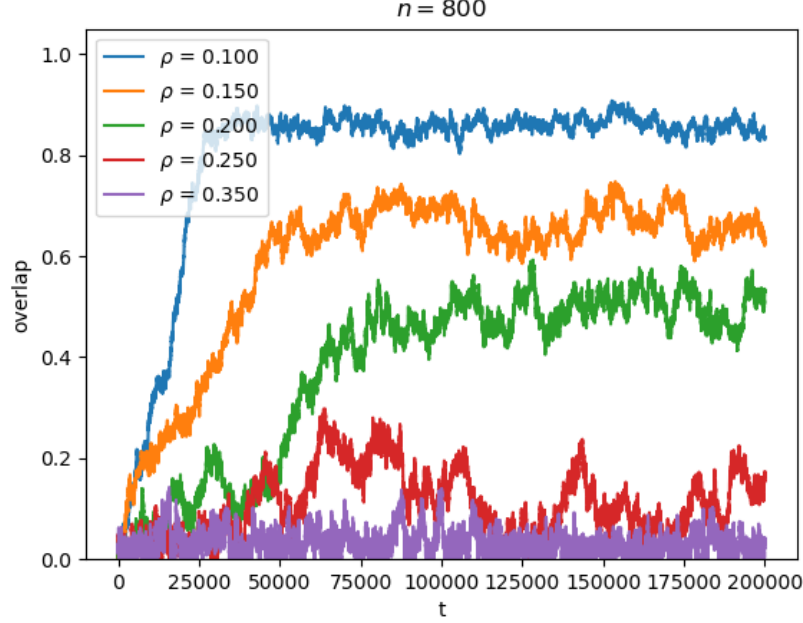


Figure 1: $Q(s_t)$. Overlaps time evolution for different ρ

where

$$\int_0^1 f(p_E) dp_E = \int_0^1 p_E^{1+\sum_{i<j} \mathbf{1}(Y_{ij}=s_i s_j) + \sum_{i<j} \mathbf{1}(Y_{ij}=-s_i s_j)} (1-p_E)^{\sum_{i<j} \mathbf{1}(Y_{ij}=s_i s_j)} dp_E$$

Substituting the final expression of both the prior and the likelihood in the posterior one finally obtains the following proportionality relation:

$$P(s|Y, \rho) \propto \prod_i \mathbf{1}(s_i = \pm 1) (1 - \rho)^{\sum_{i<j} \mathbf{1}(Y_{ij}=s_i s_j)} \rho^{\sum_{i<j} \mathbf{1}(Y_{ij}=-s_i s_j)}$$

2 Question 5

Once the Metropolis-Hastings algorithm has been implemented one can look at the evolution of the overlap during the simulation, as shown in Fig.1. The Metropolis-Hastings algorithm implemented samples spin configurations according to the posterior probability distribution $P(s|Y, \rho)$ when the MCMC has reached equilibrium. This stationary state is reached after a certain number of steps T_{therm} that depends on the value of ρ . There are mainly two ways to qualitatively estimate the value of T_{therm} . The first is to look at the behaviour of our observable $Q(s_t)$ along time, i.e. the steps of the MCMC, and look when

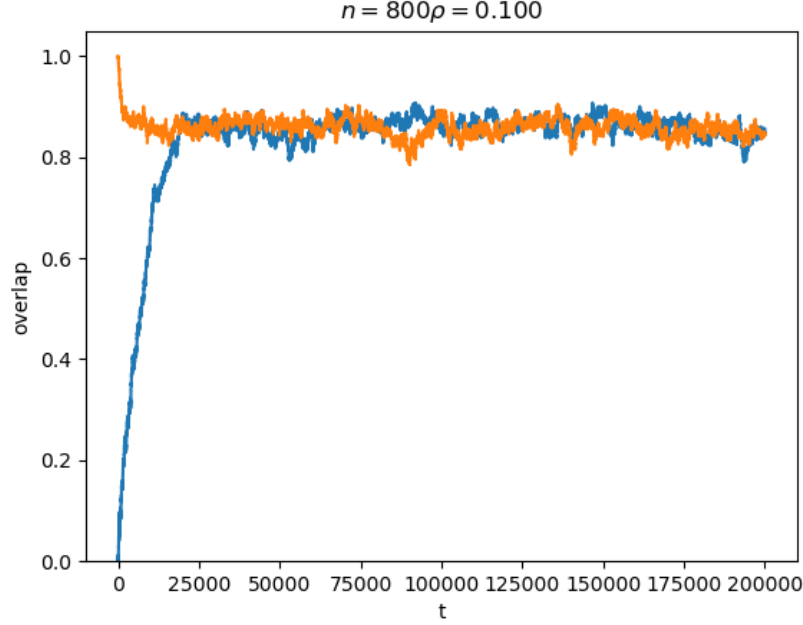


Figure 2: estimation of T_{therm} at $\rho = 0.1$

it becomes stationary. The other method is to run a second MCMC initialized at the true configuration s^* and run it in parallel with the usual MCMC initialized at a random configuration s^0 , independent from s^* . In this case T_{therm} is the time at which the two Markov Chains have approximately the same value. Of course this second method requires the knowledge of s^* , which is usually unknown in real problems. Anyway in this framework s^* is known and so this second approach has been chosen. As we can see in Fig.2, Fig.3 and Fig.4 the value of T_{therm} varies as a function of ρ . For $\rho = 0.1$, T_{therm} corresponds roughly to 25000 steps of the Markov Chain. For $\rho = 0.35$ T_{therm} is even lower. At $\rho = 0.23$, around the phase transition critical point, the two chains seem to be not totally stationary as far from this point. T_{therm} value increases in this case. Starting from the configurations sampled from the "thermalized" MCMC it is possible to build an approximation of the Bayes-optimal estimator s^{BO} . This estimator maximises the overlap

$$Q(\hat{s}) = \left| \frac{1}{n} s \cdot \hat{s} \right|$$

which is equivalent to finding s_i^{BO} such that

$$s_i^{BO} = \arg \max_{s=\pm 1} P(s_i = s | Y, \rho)$$

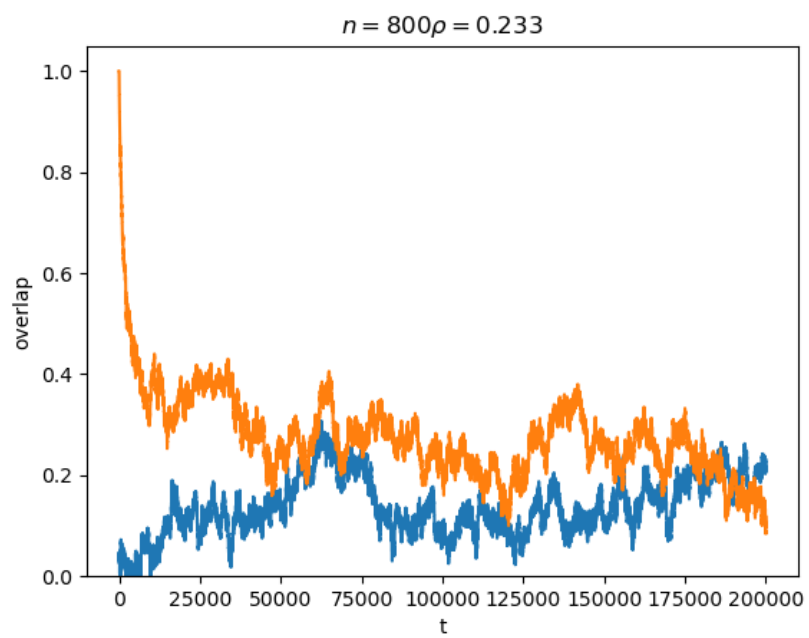


Figure 3: estimation of T_{therm} at $\rho = 0.23$

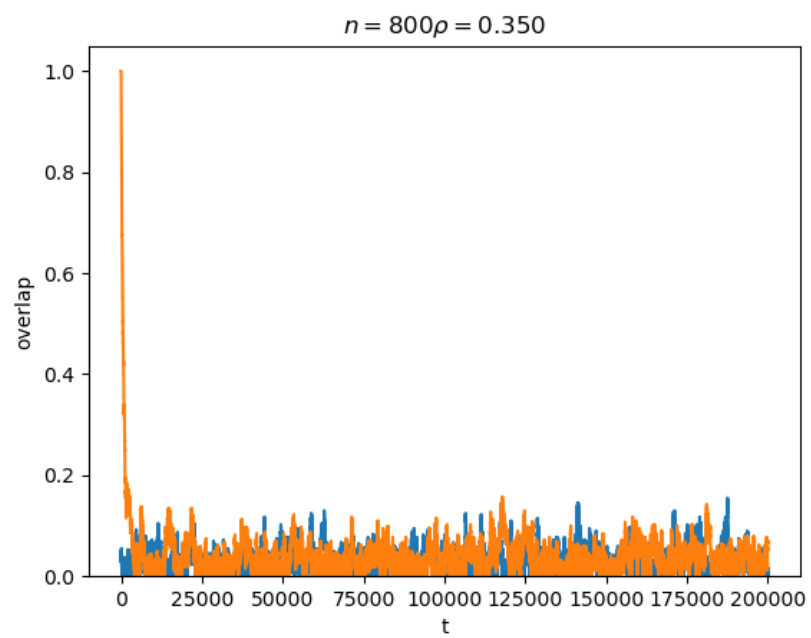


Figure 4: estimation of T_{therm} at $\rho = 0.35$

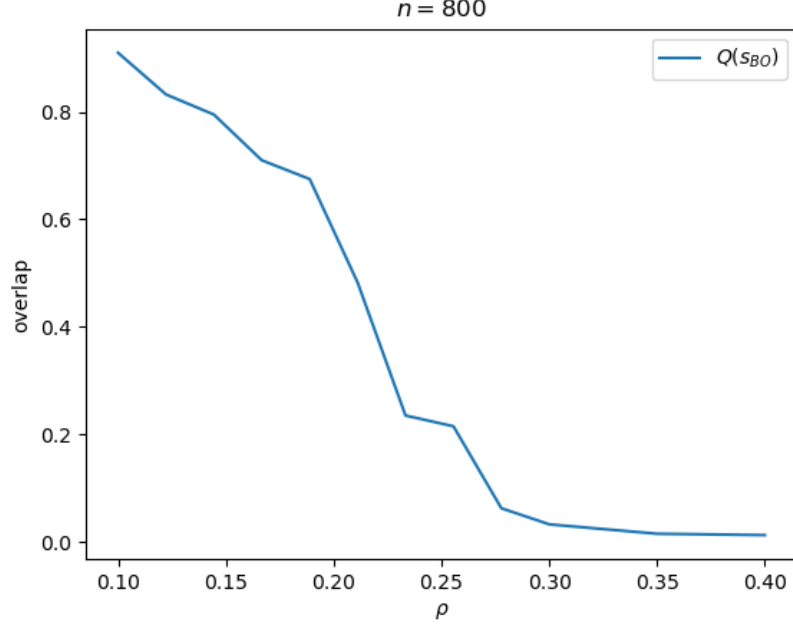


Figure 5: $Q(s_{BO})$. Phase transition observed for the overlap computed with the Bayes Optimal estimator s_{BO} , as a function of ρ

that can be rewritten as

$$s_i^{BO} = \text{sign}\left(\sum_{s_i=\pm 1} (s_i P(s_i|Y, \rho))\right)$$

So in order to find s_i^{BO} one has to compute the expected value of the spin s_i and then take the sign of it. The problem is that in order to compute the marginal $P(s_i|Y, \rho)$, one has to compute a sum over all the configurations of the other spins, which grows exponentially with the number of spins. Anyway it is possible to find an approximated value of s_i^{BO} for each spin by using the samples created by the MCMC once in equilibrium. In fact we can compute s_i^{BO} by taking the empirical average of each spin over the equilibrium configurations sampled by the markov Chain and then taking its sign. With s^{BO} it is possible to compute the overlap $Q(s^{BO})$ and it is possible to see a phase transition by plotting this observable as a function of the parameter ρ , as shown in Fig.5. The phase transition takes place around $\rho = 0.25$, where the value of the overlaps drops to zero.

Question 6

In Fig.6 the behaviours of $Q(s_{PCA})$ and $Q(s_{BO})$ are shown as functions of the parameter p_E and at fixed $\rho = 0.25$. As expected we see that the value of both the overlaps starts increasing around the value $p_E = 0.005$ which correspond to $\frac{4}{n}$. This is due to the fact that for p_E larger than 0.005 the critical value of ρ increases and therefore $\rho = 0.25$ will still correspond to high values of the overlap. In Fig.7 the same behaviours are showed by performing a mobile average on the values of the overlaps. Here is clear that the for values of $p_E > \frac{4}{n} \approx 0.005$ the $Q(s_{BO})$ has better performances than the $Q(s_{PCA})$.

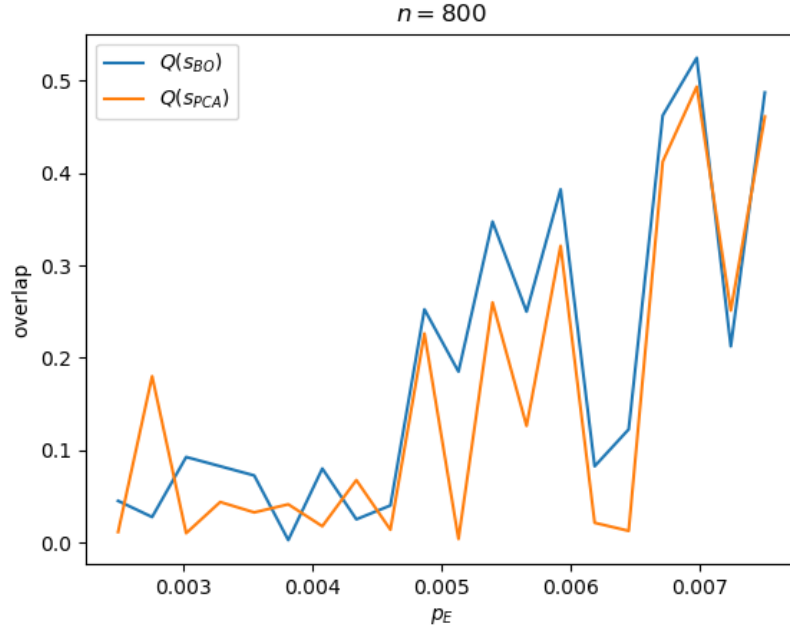


Figure 6: comparison of $Q(s_{PCA})$ and $Q(s_{BO})$ at $\rho = 0.25$

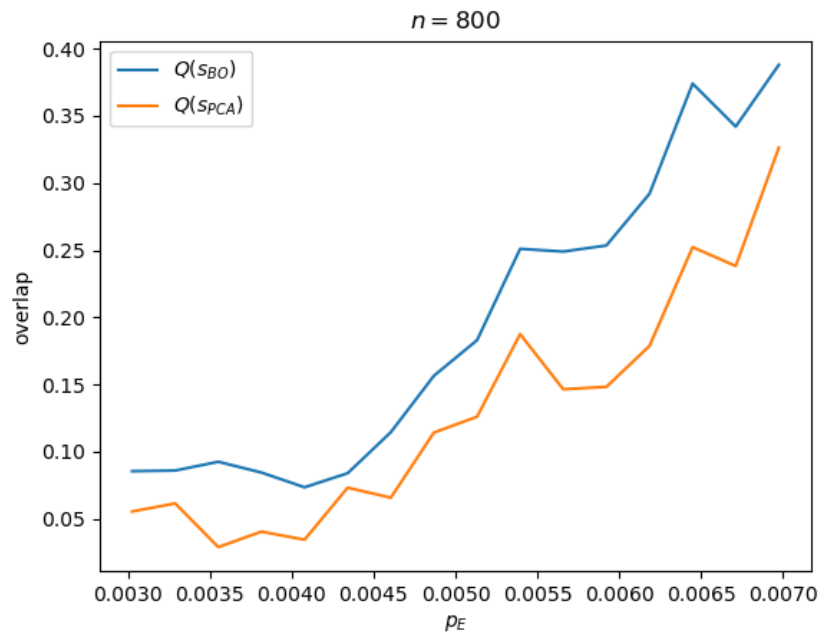


Figure 7: comparison of $Q(s_{PCA})$ and $Q(s_{BO})$ at $\rho = 0.25$ by performing mobile average between 5 near values